VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of the UGC Act, 1956)

Reg. No. : 22BAI1266

Final Assessment Test(FAT) - Nov/Dec 2024

Programme	B.Tech.	Semester	Fall Semester 2024-25
Course Code	BCSE301L	Faculty Name	Prof. Rama Prabha
Course Title	Software Engineering	Slot	G1+TG1
		Class Nbr	CH2024250100970
Time	3 hours	Max. Marks	100

General Instructions

- Write only Register Number in the Question Paper where space is provided (right-side at the top) & do not write any other details.

Course Outcomes

1. Apply and assess the principles of various process models for the software development.
2. Demonstrate various software project management activities that include planning , Estimations, Risk assessment and Configuration Management
3. Perform Requirements modelling and apply appropriate design and testing heuristics to produce quality software systems.
4. Demonstrate the complete Software life cycle activities from requirements analysis to maintenance using the modern tools and techniques.
5. Escalate the use of various standards and metrics in evaluating the process and product.

Section - I

Answer all Questions (1 × 10 Marks)

*M - Marks

Q.No	Question	*M	CO	BL
01.	An Asset Management Company (AMC) implements an Agile methodology to develop the Investor Self-Service Application (ISSA) portal. This portal aims to offer investors real-time access to their customer portfolios, enabling continuous feedback and iterative improvements. a. List different phases of agile methodologies and evaluate the effectiveness of each phase of an agile model in the development process of ISSA (5 Marks) b. Identify and illustrate the phases of Extreme Programming (XP) that significantly enhance the testing quality for an Integrated Student Support Application (ISSA). Include a clear, labelled diagram to support your answer (5 marks)	10	1	2

Section - II

Answer all Questions (3 × 20 Marks)

*M - Marks

Q.No	Question	*M	CO	BL
02.	A charity trust has decided to design and implement an online platform to enhance communication and engagement with its clients and donors.	20	2,3	3

a. Prepare a Software Requirements Specification (SRS) document for the above scenario in a standard SRS template. The SRS should include key elements such as functional and non-functional requirements, user roles, and interactions, along with specific platform features that enhance the trust's communication and engagement with its clients and donors. (7 Marks)
b. Construct a Work Breakdown Structure (WBS) for the above scenario. (6 marks)
c. Propose a risk management plan that includes risk identification, assessment, mitigation strategies, and monitoring procedures to ensure platform reliability and data security for the above scenario with a neat sketch. (7 marks)

03.	You are tasked with designing a Course Registration System (CRS) for a university. a. Draw a class diagram for a university CRS and show the relationships between each class, including associations, aggregations, and inheritance for the same. (8 marks) b. Draw a use case diagram for a university's CRS, showing how students, faculty, and administrators interact with it for tasks like enrolling in courses and checking availability. (5 marks) c. Identify the different types of User Interfaces and design principles that would be essential for the CRS. (7 marks)	20	3	4
04.	You are the Quality Assurance (QA) lead for a newly developed mobile application aimed at delivering educational resources to university students. The app includes features such as video streaming, push notifications, offline access, and a user-friendly interface. As the QA lead, it's crucial to identify and conduct specific types of testing to ensure the app's performance, usability, and reliability across various mobile devices and operating systems. a. Identify the key types of testing essential for the given mobile application and explain why each type is important. (6 Marks) b. Considering the critical importance of real-time order processing, describe your approach for conducting regression testing to ensure that existing functionalities remain unaffected by system updates. (8 Marks) c. Classify the Software Architecture and Software Design for the given scenario, with a neat diagram. (6 Marks)	20	3,4	4

Section - III
Answer all Questions (2 × 15 Marks)

*M - Marks

Q.No	Question	*M	CO	BL
05.	An Inventory Management System (IMS) aims to enhance organizational operations by modernizing legacy systems and integrating advanced technologies. Success relies on continuous improvement, active stakeholder involvement, and data-driven decision-making to streamline processes and drive growth. a. Implement re-engineering and reverse engineering processes for the IMS (7 Marks) b. Demonstrate how Total Quality Management (TQM) principles can be applied to streamline the Inventory Management System. (8 marks)	15	4,5	3
06.	Imagine you are a software engineer at a growing technology firm tasked with improving software development processes. To enhance quality and ensure consistent, reliable project outcomes, the firm has adopted the Capability Maturity Model (CMM). a. Illustrate the stages of the Capability Maturity Model (CMM) in the context of enhancing quality within a software development process for the given scenario. (8 marks) b. Design a Software Process Improvement (SPI) framework for the technology firm (7 marks)	15	5	2

BL-Bloom's Taxonomy Levels - (1.Remembering, 2.Understanding, 3.Applying, 4.Analysing, 5.Evaluating, 6.Creating)